STRATEGIC PARTNERSHIP OFFER

Open Bioengineering & AI Safety Architectures

"Althea" (CRISPR-TTP) + "AMAGI" (Hardware-Enforced AI Safety)

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Commercial Enquiries: Subject to S-APL (Strict Architectural Principle License)

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Legal Entities: LLC "Althea" (Russia) / AGIAM Technologies Ltd (UK, inactive shell)

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1. Executive Summary

We present two complementary, fully open architectures that together form a complete ecosystem for **trustworthy, life-critical autonomous systems**:

1.1 "Althea" – CRISPR-TTP Bioengineering Platform

A fully open (CC0) bioengineering architecture for the treatment of fusion-driven cancers, with an initial focus on **Ewing Sarcoma** – the second most common bone cancer in children and young adults.

- **Core technologies:**
  - **CRISPR-Cas9 genome editing** of autologous dendritic cells (>94% efficiency, 17+9 MHC binders);
  - **Temporally Programmed Delivery (TTP)** via MR-guided Focused Ultrasound (FUS) activating HOF nanoparticles (27 ± 3 nm, 1–2 mm spatial resolution);
  - **AI Co-pilot** for real-time personalization, off-target risk mitigation, and closed-loop therapeutic control (89.3% sgRNA prediction accuracy).

- **Core publications:**
  - Official publication: **Frontiers in Genetics** (2026) – DOI: [10.3389/fgene.2026.1727708](https://doi.org/10.3389/fgene.2026.1727708)
  - Full architectural specification v4.0 (preprint) – DOI: [10.5281/zenodo.17780623](https://doi.org/10.5281/zenodo.17780623)
  - Clinical protocol SOP v2.0 – DOI: [10.5281/zenodo.17676423](https://doi.org/10.5281/zenodo.17676423)
  - TTP v3.0 (delivery architecture) – DOI: [10.5281/zenodo.17237907](https://doi.org/10.5281/zenodo.17237907)
  - CRISPR Autovaccination v3.0 – DOI: [10.5281/zenodo.17246573](https://doi.org/10.5281/zenodo.17246573)

1.2 "Amagi" – Hardware-Enforced AI Safety Architecture

A reference architecture designed to provide structural safety assurances for high-risk AI systems, transforming regulatory compliance from a procedural burden into architectural guarantees.

- **Core principles:**
  - **Six immutable invariants** (Physical Non-Bypassability, Irreducible TCB, Separation of Control, Immutability of Policy, Complete Mediation, Complete Audit Integrity);
  - **TTP Triad** (Trigger → Programmable Delay → Localized Activation) as a universal safety primitive;
  - **Hardware components:** SCC (Security Council Core), NIPU (Neuro-Immune Processing Unit), MEM (Self-Protecting Memory), Amagi-Q (post-quantum crypto);
  - **Alignment** with EU AI Act, NIST AI RMF 1.0, ISO 26262 (ASIL-D) frameworks.
- **Core publications:**
  - Amagi Framework v1.4 – DOI: [10.5281/zenodo.17849595](https://doi.org/10.5281/zenodo.17849595)
  - Amagi Framework v1.3 – DOI: [10.5281/zenodo.17696607](https://doi.org/10.5281/zenodo.17696607)
  - Amagi Framework v1.0 – DOI: [10.5281/zenodo.17679248](https://doi.org/10.5281/zenodo.17679248)
  - Amagi-NIPU v1.0 (processor) – DOI: [10.5281/zenodo.17682668](https://doi.org/10.5281/zenodo.17682668)
  - Amagi-MEM v1.0 (memory) – DOI: [10.5281/zenodo.17679327](https://doi.org/10.5281/zenodo.17679327)
  - NIST AI RMF Profile – DOI: [10.5281/zenodo.17849725](https://doi.org/10.5281/zenodo.17849725)
  - AMAGI OPEN SPECIFICATION V1.0 – DOI: [10.5281/zenodo.17861024](https://doi.org/10.5281/zenodo.17861024)
  - AGIAM Founding Charter – DOI: [10.5281/zenodo.17719569](https://doi.org/10.5281/zenodo.17719569)
  - Trade Secrets Notice – DOI: [10.5281/zenodo.17886699](https://doi.org/10.5281/zenodo.17886699)
  - Amagi Architectural Class Definition – DOI: [10.5281/zenodo.17938573](https://doi.org/10.5281/zenodo.17938573)

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2. Current Status & Achievements

2.1 "Althea" (CRISPR-TTP)

Area	Status	DOI Reference
Peer-reviewed publication	Frontiers in Genetics (Q1, CiteScore 6.2) – Feb 11, 2026	[10.3389/fgene.2026.1727708](https://doi.org/10.3389/fgene.2026.1727708)

Area	Status	DOI Reference
Regulatory dialogue	EMA: ATMP classification determined (Feb 11, 2026)	–
Regulatory dialogue	FDA CBER: INTERACT meeting submitted (Feb 18, 2026)	–
Regulatory dialogue	Russian MoH: Expert evaluation completed (Feb 19, 2026)	–
Clinical protocol	21-day GMP-compatible SOP v2.0	10.5281/zenodo.17676423
Foundational architectures	TTP v3.0 (delivery)	10.5281/zenodo.17237907
Foundational architectures	CRISPR Autovaccination v3.0	10.5281/zenodo.17246573

2.2 "Amagi" (AI Safety)

Area	Status	DOI Reference
Scientific publications	Amagi Framework v1.4	10.5281/zenodo.17849595
Scientific publications	Amagi Framework v1.3	10.5281/zenodo.17696607
Scientific publications	Amagi Framework v1.0	10.5281/zenodo.17679248
Technical specifications	Amagi-NIPU v1.0 (processor)	10.5281/zenodo.17682668
Technical specifications	Amagi-MEM v1.0 (memory)	10.5281/zenodo.17679327
Regulatory alignment	NIST AI RMF Profile (Target State)	10.5281/zenodo.17849725
Open standard	AMAGI OPEN SPECIFICATION V1.0	10.5281/zenodo.17861024
Legal foundation	AGIAM Founding Charter	10.5281/zenodo.17719569
	Trade Secrets Notice	10.5281/zenodo.17886699
	Amagi Architectural Class Definition	10.5281/zenodo.17938573

3. The Unmet Need – Why Both Architectures Are Necessary

3.1 Medical Challenge

Ewing Sarcoma is a devastating paediatric malignancy. Despite multimodal therapy, **5-year survival for metastatic disease remains below 30%**. The pathognomonic **EWSR1-FLI1 fusion oncogene** is an ideal tumour-specific target, yet it has proven undruggable by conventional pharmacology. The tumour microenvironment is immunologically "cold", with downregulated HLA-I and high Treg infiltration.

3.2 AI Safety Challenge

Modern AI systems controlling life-critical functions are inherently unpredictable. Traditional software-based safety measures can be bypassed, creating unacceptable risks. Regulators (EU AI Act, NIST, FDA) now demand structural guarantees, not just procedural promises.

Althea addresses the biological complexity. Amagi addresses the computational risk. Together, they enable a future where autonomous therapeutic systems are both effective and designed for trustworthiness.

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4. What We Have Already Achieved

4.1 Technical Completeness

- **Althea:** CRISPR-DC, TTP v4.0, AI Co-pilot, 21-day GMP protocol. Full specification, peer-reviewed.
- **Amagi:** SCC, NIPU, MEM, ICG, TDF, Amagi-Q, 6 core principles. Full specifications, open standard.

4.2 Legal & Organisational Foundation

- **LLC "Althea" (Russia)** – operational entity for medical development.
- **AGIAM Technologies Ltd (UK)** – international licensing and certification shell.
- **S-APL (Strict Architectural Principle License)** – protects architectural integrity.
- **Trade Secrets** – Hard IP Suite (GDSII, netlist) protected under DTSA, EU Directive 2016/943.
- **Prior Art** – core principles publicly disclosed on Zenodo.

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5. What We Are Looking For

We invite partners to join us in translating both architectural blueprints into experimental and clinical reality.

5.1 Preferred Areas of Collaboration

Area	Description	Relevant Architecture
Preclinical Validation	CROs or academic labs for efficacy/safety studies.	Althea
GMP Manufacturing	Centres for DC production and CRISPR editing.	Althea
Clinical Trials	Hospitals for early-phase trials (Phase I/II).	Althea
AI Co-pilot Development	Partners to co-develop the "Althea Copilot".	Althea + Amagi
Hardware Implementation	Semiconductor companies for ASIC/FPGA IP cores.	Amagi
Regulatory & Market	Consultants for EMA/FDA/ATMP/AI certification.	Both
Funding & Grants	Co-application for Horizon Europe, EIC, etc.	Both

5.2 Flexible Partnership Formats

- Scientific collaboration (joint publications);
- Service agreements (CRO, manufacturing);
- Grant consortia;
- Technology/IP core licensing (S-APL);
- Equity-free sponsorship (philanthropic or corporate);
- Advisory board membership.

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6. Why Partner With Us?

- **Open by design:** No IP battles – all knowledge is freely available as prior art with DOIs.
- **Regulator-ready:** Already engaged with EMA (Feb 11), FDA (Feb 18), and Russian MoH (Feb 19).
- **Standards-aligned:** Amagi provides a complete NIST AI RMF Profile and alignment with EU AI Act.

- **Modular & adaptable:** Althea can be repurposed for other malignancies; Amagi applies to any high-risk AI.
- **Ethically grounded:** Dedicated to the memory of Althea (Irina Neboga) – a humanitarian mission.
- **Cost-effective:** Open specifications reduce R&D duplication; transparent, mission-driven model.
- **AI safety by design:** Amagi provides structural safety assurances aligned with global regulations.
- **Complete ecosystem:** From therapeutic mechanism to hardware-enforced AI safety.

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7. Complete List of Published Works

7.1 Althea / CRISPR-TTP Series

1. Official Publication (v4.0) – 10.3389/fgene.2026.1727708
2. Genomic Innovation v4.0 Preprint – 10.5281/zenodo.17780623
3. SOP Althea v2.0 – 10.5281/zenodo.17676423
4. Clinical Protocol v1.0 – 10.5281/zenodo.17537570
5. Genomic Innovation v3.0 – 10.5281/zenodo.17526578
6. Genomic Innovation v2.0 – 10.5281/zenodo.17308402
7. Integrated Architecture v1 – 10.5281/zenodo.17308401
8. URGENT MEDICAL CASE – 10.5281/zenodo.17247084
9. CRISPR Autovaccination v3.0 – 10.5281/zenodo.17246573
10. TTP v3.0 – 10.5281/zenodo.17237907

7.2 Amagi / AGIAM Series

1. Amagi Architectural Class Definition – 10.5281/zenodo.17938573
2. Trade Secrets Notice – 10.5281/zenodo.17886699
3. AMAGI OPEN SPECIFICATION V1.0 – 10.5281/zenodo.17861024
4. NIST AI RMF Profile – 10.5281/zenodo.17849725
5. Amagi Framework v1.4 – 10.5281/zenodo.17849595
6. AGIAM Founding Charter – 10.5281/zenodo.17719569
7. Amagi Framework v1.3 – 10.5281/zenodo.17696607
8. Amagi-NIPU v1.0 – 10.5281/zenodo.17682668
9. Amagi-MEM v1.0 – 10.5281/zenodo.17679327
10. Amagi Framework v1.0 – 10.5281/zenodo.17679248

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8. Next Steps

1. **Initial contact:** Send an expression of interest to **lexintrud@gmail.com** (subject: "Partnership Inquiry – Althea/Amagi").
2. **NDA & information exchange** (if required).
3. **Discussion of specific collaboration format.**
4. **Formal agreement** (partnership, service, or licensing contract).

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9. Contact & Legal Information

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Legal Entities:

- **LLC "Althea" (Russia):** Active (INN 4217214140, OGRN 1264200000900)
- **AGIAM Technologies Ltd (UK):** Inactive shell, ready for activation.

Licensing:

- **Althea (medical):** CC0 1.0 Universal – freely available for any purpose.
- **Amagi (AI safety):** CC BY-NC 4.0; S-APL required for commercial use.
- **Hard IP Suite:** Protected as trade secrets; access requires NDA + S-APL.

DISCLAIMER

This document describes architectural concepts and research frameworks. It does not constitute medical advice, regulatory approval, or investment solicitation. All clinical and hardware implementations require independent validation and regulatory authorization.

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No external funding was received for the preparation of this document.

CONFLICT OF INTEREST

The author is founder of LLC "Althea" and AGIAM Technologies Ltd.

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